



Fluorescence imaging for mm-scale observation of macropore-matrix mass transfer: Calibration experiments

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ABSTRACT

Macropores are important for gas, water, and heat movement in soils. Still, the transfer of water and solutes through the interface between macropore and matrix like the walls of biopores is less understood. For mass-exchange processes and water storage in soil aggregates, relatively small sizes and volumes limit the analysis of distributed interface properties including the permeability for water or air. Photo-based imaging techniques could perhaps be used to trace the movement of sorbing chemicals at such smaller scale similarly as for larger soil profiles. The objective was to test the suitability of Na-Fluorescein (i.e., fluorescein di sodium salt) for marking small-scaled flow and transport processes through surfaces of biopore walls and coated soil aggregates and for determining water and reactive solute mass exchange.

Fluorescence imaging was carried out using fluorescence spectrometer (for aqueous solutions) and a uv/vis photo-based imaging technique for soil and intact biopore and aggregate surfaces of soil samples from the Bt-horizon of a haplic Luvisol. For calibration, a linear relation between fluorescence intensity and initial Na-Fluorescein concentrations could be used for aqueous solutions up to a concentration of 100 mg L^{-1} Na-Fluorescein. The fluorescence intensities of homogenized soil samples were used to determine the amount of sorbed dye at equilibrium.

After calibration, the photo-based imaging technique allows determining mm-scale maps of dye concentrations at intact soil surfaces in both, the dissolved and sorbed fraction of the dye. Uncertainties are caused mainly by effects of fluorescent soil organic carbon that is heterogeneously distributed at biopore wall surfaces. The results suggest that sorption of the dye mostly occurs in the soil region next to the macropore surface. A few high concentration spots at greater distances could be an indication for rapid tracer movement along root channels suggesting occurrence of non-uniform flow and transport even at the scale of soil clods and aggregates.

The method seems promising for a small-scale analysis of macropore-matrix exchange of reactive solutes.

1. Introduction

Na-Fluorescein (i.e., Fluorescein sodium ($\text{C}_{20}\text{H}_{10}\text{Na}_2\text{O}_5$, Falbe and Regitz, 1998)) has been used as dye tracer in hydrological experiments for decades because it is the most intensively colored substance known (Käss, 1998). Today, dyes like Na-Fluorescein are widely used to stain flow paths of water and solutes in soils (e.g., Aeby et al., 2001; Flury and Fühler, 1995; Gerke et al., 2008, 2015; Vanderborght et al., 2002; Soto-Gómez et al., 2018). More recently, it has been used to determine the dispersivity coefficient (Citarella et al., 2015), the sorption properties of soils (Gerke et al., 2008) and to visualize the vaporization plane within porous media (Weiss et al., 2018). Smart and Laidlaw (1977) assumed that Fluorescein is sorptive to soils at moderately basic

to neutral pH-values; contrastingly, Käss (1998) found only small but Gerke et al. (2008) larger sorption by soils. Differences in the observed sorption behavior are caused by the Fluorescein molecule, which is a cation ($\text{pK}_1 = 1.95$) when protonated, a neutral molecule ($\text{pK}_2 = 5.05$), or result in mono- or divalent anions ($\text{pK}_3 = 7.0$) when deprotonated (Zanker and Peter, 1958). In addition to pH-effects, sorption properties of Fluorescein are also affected by soil texture and soil organic carbon. The pH effect on Fluorescein's structure is important because the ability to fluorescence of Fluorescein is mostly related to the presence of conjugated C–C double bonds (Falbe and Regitz, 1998). Such conjugation of the double bonds is most widespread across the Fluorescein's molecular structure in case of the mono- or divalent dissociated anionic species (Falbe and Regitz, 1998) such that fluorescence will be

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lost in case of protonation (i.e., in case Fluorescein is applied to acidic soils). Adsorption on soil minerals or soil organic matter may also affect the conjugated system resulting in a loss of fluorescence.

Compared to the bulk of the soil matrix, macropores show differences in structural (Haas and Horn, 2018) and physico-chemical properties, such as the organic matter (OM) content and composition (Ellerbrock et al., 2009), in terms of hydrophilic and hydrophobic OM functional groups that affect the local soil water repellency (Leue et al., 2010, 2016; Haas et al., 2018). Soil macropores can be grouped in biopores (i.e., earth worm burrows, root channels) and shrinkage cracks. The surfaces of these macropores are different as compared to the bulk soil with respect to morphological features (Haas and Horn, 2018) and ecological functions (Zhang et al., 2018). We hypothesize that (I) the sorption of solutes to the walls of macropores differs from the soil matrix due to alterations of pore structural and physico-chemical properties and (II) the macropore-matrix exchange flux is spatially heterogeneous at the local scale, especially when considering reactive solutes. For analyzing processes at the macropore-matrix interface (Gerke et al., 2013a), imaging methods are needed that account for the small-scaled and highly dynamic interactions of biological, chemical, and physical processes in such soil regions. In particular, a technique for the temporally variable identification of effects of aggregate-scale spatial heterogeneity of macropore surface properties is needed that can be used for experimental determination and parameterization of the solute mass transfer in two-domain flow and transport modeling of preferential flow (e.g., Gerke, 2006).

The objectives of this paper are (i) to test if Na-Fluorescein is applicable to mark small-scaled flow and transport processes through surfaces of biopore walls and coated soil aggregates, (ii) to establish relationships between Na-Fluorescein tracer concentration *versus* fluorescence-intensity for determination of macropore wall sorption.

Batch experiments with Na-Fluorescein were performed to determine the sorption properties of the dye on homogenized soil and transport experiments based on spraying Na-Fluorescein tracer on intact soil blocks with biopore walls and crack surfaces from a structured Luvisol Bt-Horizon to estimate mm-scale water and solute movement through the macropore-matrix interfaces.

2. Material and methods

2.1. Soil and sample preparation

The loam to sandy loam soil (165 g kg⁻¹ clay, 350 g kg⁻¹ silt, 485 g kg⁻¹ sand, 3.1 g kg⁻¹ soil organic carbon) was the same as used in previous studies (Leue et al., 2013); samples were from the clay-enriched Bt horizon (located in 0.4–0.6 m depths) of an arable Haplic Luvisol (IUSS Working Group WRB, 2006) developed on glacial till, located at CarboZALF-D experimental field, Holzendorf, near Prenzlau in northeastern Germany. For the calibration of imaged color and grayscale values in relation to Na-Fluorescein concentrations, soil was sieved to pass 0.63 mm and mixed with aqueous solutions (1:2.5 w/w, soil:water) of defined initial Na-Fluorescein concentrations (c_i) of 0, 25, 50, 75, 125 and 150 mg L⁻¹ to obtain a soil slurry ($n = 3$ for each c_i). While for one sample set no buffer was added, a second sample set was buffered using 0.05 M K₂HPO₄/KH₂PO₄ to a pH value of 7.05. The sufficiency of the amount of the buffer was verified by measuring the pH-values of the suspensions using the pH-meter inoLab® pH 7310 (Xylem Analytics Germany Sales GmbH & Co. KG, Weilheim, Germany). The soil slurries were shaken in centrifuge cups using IKA Orbital Rotary Shaker KS 501 D (IKA®-Werke GmbH & Co. KG, Staufen, Germany) with 250 rpm for 48 h (Vanderborght et al., 2002), then centrifuged for 0.5 h with 4000 rpm using Centrifuge 5804 (Eppendorf AG, Hamburg, Germany), and finally filtrated using a polyamid filter, namely, NL17 (GE Healthcare GmbH, Solingen, Germany) with pore diameter ≤ 0.45 mm. The liquid phase was diluted (1:100 v/v sample:water) and used for fluorescence spectroscopy (see Section 2.3). The remaining

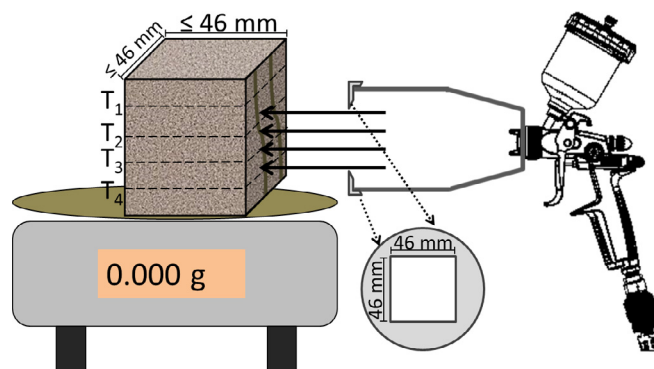


Fig. 1. Schematic drawing of the experimental set-up for tracer application. The soil sample is placed on an electrical balance while it is sprayed with a Na-Fluorescein solution using an air brush gun (Minijet 3000B). The applied pressure of the air brush of 0.2 MPa was generated by using a compressor.

water-saturated soil was filled in excess into white plastic caps (25 mm in diameter, 3 mm in height) using a miniature spatula. The lapping soil was slightly pressed into the cap and, afterwards, smoothed by brushing carefully over the opening of the cap. The resulting sample surfaces were plane, smooth and slightly covered with a film of water when used for fluorescence imaging.

One intact larger soil aggregate with about 5 cm edge length of the structured soil used to test the applicability of the approach was excavated from the same Bt horizon of that arable Luvisol. The soil aggregate was air-dried under laboratory conditions and sealed at all sides with tinfoil and adhesive tape. Afterwards, the cover at the surface of interest (SOI) that received the sprayed tracer solution (≤ 46 mm in width and height) was removed. In the following wetting procedure, an aqueous solution with a defined initial Na-Fluorescein concentration of 100 mg L⁻¹ (buffered to a pH of 7.05 using 0.05 M K₂HPO₄/KH₂PO₄) was sprayed horizontally at the side face of the sample with the intact coatings (Fig. 1). For spraying, an air brush gun was used while the sample was placed on an electrical balance for controlling the applied spraying rate of 0.9 mg s⁻¹. The plan-view fluorescence imaging of the plan view soil surface was carried out after horizontally slicing the soil aggregate perpendicular to the intact coated SOI (Fig. 1). Slicing in about 10 mm vertical steps was done after spraying for 1, 2, 3, and 4 h as follows: The spraying was stopped to move the sample from the electrical balance. The sealing was removed from the 1–2 uppermost centimeters of the sample. Following natural fissures, individual smaller aggregates (≤ 10 mm) were manually sheared off from the top of the sample by using a mini-spatula. From the prepared horizontal plan view surface, the fluorescence image was taken and immediately afterwards, the plan surface was covered again before the wetting procedure continued until the next plan view was sliced and a fluorescence image was taken. The sample sides were covered to maintain the horizontal water movement by preventing evaporation from and wetting of surfaces other than the SOI.

2.2. Fluorescence imaging

For the fluorescence imaging, the dark hood DH-50 (Biostep GmbH, Burkhardtshof, Germany) and digital camera EOS 700 (Canon, Tokio, Japan) with objective EI 28 mm f/1.8 USM (Canon, Tokio, Japan) was used. Both, soil slurries in caps and the sliced intact soil sample surface were centered within the DH-50 immediately before 8-bit images were taken in two steps: First, integration time of images was set to 0.250 s for photos of samples illuminated with light in the visible spectrum; second, image integration time was set to 30 s under illumination with UV radiation. In both cases, the diaphragm of the camera was set to “5”, using the software biostep argusX1, Version 7.14.22 (Biostep GmbH, Burkhardtshof, Germany).

Imaging was performed in color mode. Here, three values representing red, green, blue are obtained with the camera, representing weighted emission intensities of defined wavelengths or “tristimulus values” according to Trussell et al. (2005). The weighted individual color values were linked to the grayscale (GS) value (IEC 61966-2-1, 1999) as:

$$GS = a_r r + a_g g + a_b b \quad (1)$$

where r denotes the color value for red, g for green, and b for blue (RGB-values) and $a_r = 0.2126$, $a_g = 0.7152$, and $a_b = 0.0722$ are dimensionless coefficients.

Both, the RGB and the GS values were tested as a measure of fluorescence intensity. The repeatability of color values (indicated by low standard deviations) and calculated grayscale values within each image was tested by dividing each image in 169 sub-images consisting of 169 pixels each (13×13).

2.3. Fluorescence spectroscopy and data processing

The uv/vis spectrometer Aqualog (Jobin Yvon Technologies, Horiba Scientific, Kyoto, Japan) was used to determine excitation emission matrices (EEM) of liquid samples from the aqueous solutions used to saturate the soil slurries and from the aqueous liquid collected from soil slurries by centrifugation and filtration (see Section 2.1). The EEM is a 3D scan, resulting in a contour plot of excitation wavelength vs. emission wavelength vs. fluorescence intensity. Aqueous solutions were transferred into 0.0025 L volume UV-stable and -transparent disposable cuvettes, 4 sided clear (Brand GmbH + Co. KG, Wertheim, Germany).

Data evaluation started with the selection of excitation wavelengths (λ_{Ex}) between 310 and 530 nm in 3 nm increments. For each λ_{Ex} , the global maximum in the peak emission intensity was identified for all recorded emission wavelengths (λ_{Em}) between 212 and 620 nm of aqueous solutions with defined initial Na-Fluorescein concentrations (c_i). In order to account for the shift in the emission spectra, cumulated emission intensities in ranges between 3 and 40 nm around the emission peak were tested for finding optimal relationships between the fluorescence intensity and the Na-Fluorescein concentration. Both, the maximum peak emission intensity and the cumulated emission intensities were regression-fitted to the measured concentrations by using linear and logarithmic functions. For describing the calibration relationships between emission intensity and Na-Fluorescein concentration, the fit with the highest R^2 was chosen (i.e., for single peak intensity and λ_{Ex} of 420 nm). The relationship was used to determine the concentration of Na-Fluorescein in the equilibrium (c_{eq}) solution of the soil-water mixtures (after centrifugation). The concentration of Na-Fluorescein sorbed to soil (c_s) was determined by assuming $c_i = c_s + c_{\text{eq}}$. The RGB-values from imaging the soil samples were combined to grayscale values and related to the c_{eq} by assuming that the total Fluorescence intensity was caused by Na-Fluorescein in solution.

Data analysis was carried out with the statistical software R (R Development Core Team, 2018). Python2.7 scripting language (Python Software Foundation, 2019) and additional python packages (PIL (Wiredfool et al., 2016), NumPy (NumPy developers, 2018) and cv2 (Bradski, 2009)) were used for image processing.

3. Results and discussions

3.1. Fluorescence intensity versus Na-fluorescein concentration

With the applied settings, the pixel size was $69 \mu\text{m}$, which is close to the diameter of pores that are drained at field capacity ($\psi_m = -6 \text{ kPa}$). Fluorescence intensity obtained from imaging in the dark hood increased with Na-Fluorescein concentration, and differed for buffered and unbuffered soil slurries (samples in caps). The example pictures (Fig. 2) revealed that the fluorescence intensity was spatially

heterogeneous especially visible for samples with concentrations $\geq 2 \text{ mg L}^{-1}$. The samples without Na-Fluorescein (Fig. 2, left) are completely dark, while fluorescence intensity of samples treated with 200 mg L^{-1} Na-Fluorescein solution reached the saturation threshold of the fluorescence imaging detector as indicated by completely white pixels (Fig. 2, right). As measure for the variability within each image, values of the coefficient of variation (CV) were calculated and ranged for grayscale values between 0.9 and 29% (Table 1) for the 169 sub-images consisting of 169 pixels.

For the soil slurry samples, the effect of the pH-value on the fluorescence intensity in terms of color (Fig. 3) and grayscale (Fig. 4) values was most pronounced for the 20 mg L^{-1} Na-Fluorescein concentration. Differences between buffered and unbuffered samples were relatively small for all other Na-Fluorescein concentrations. Fluorescence intensities were slightly smaller in unbuffered as compared to buffered samples for 0 to 20 mg L^{-1} Na-Fluorescein (Fig. 4). While the values of the color channels increased with concentration from 2 to 20 mg L^{-1} for buffered solutions (Fig. 3a), the fluorescence intensity remained unchanged for unbuffered samples (Fig. 3b). The grayscale value of buffered samples increased from 2 to 20 mg L^{-1} more intensely as compared to that of the unbuffered samples (Fig. 4) indicating that for these samples, the lower detectable Na-Fluorescein concentration decreased from 20 mg L^{-1} to 2 mg L^{-1} by buffering. This is potentially caused by changes in the species of Na-Fluorescein, which are related to pH (c.f., Zanker and Peter, 1958).

3.2. Fluorescence spectroscopy and effect of excitation wavelengths

The influence of excitation wavelengths (λ_{Ex}), here between 350 nm and 490 nm, on the relationship between fluorescence intensity and initial Na-Fluorescein concentration (c_i) of aqueous solutions was relatively strong (Fig. 5). For λ_{Ex} of 420 nm, the relation was almost linear in the range of c_i between 0 and 200 mg L^{-1} , while for λ_{Ex} of 490 nm, the upper sensor range of the spectrometer was reached with a c_i of 100 mg L^{-1} Na-Fluorescein.

For aqueous solutions at relatively low Na-fluorescein concentrations (i.e. $c_i < 0.4 \text{ mg L}^{-1}$), the filtration affected the fluorescence intensity over a wide range of excitation and emission wavelengths (for example, for $\lambda_{\text{Ex}} = 489 \text{ nm}$ in Fig. 6). Here, filtration (see Section 2.1) caused a reduction in fluorescence intensity of up to 50%, and reduced the shift to the reddish. Filtration effect was found for wavelengths frequently used for fluorescein concentration determination such as 489 nm (Gerke et al., 2013b). The use of polyamide filters ($0.45 \mu\text{m}$ in diameter) in combination with $\lambda_{\text{Ex}} = 490 \text{ nm}$ was found inappropriate; we assumed that fluorescent organic compounds were separated from the filtrate by the $0.45 \mu\text{m}$ polyamid filter. To avoid this problem, we used a sedimentation method instead of filtration for the relatively small concentrations during the calibration.

The alternative approach to use fluorescence intensity at a λ_{Ex} value of 350 nm as recommended by Gerke et al. (2013a,b), could potentially be influenced by fluorescence-active organic impurities like bacteria or dissolved organic matter (DOM) (Hansen et al., 2016). Wavelengths in the range of 254–370 nm are typically used to determine the quality of organic compounds, e.g., as indicator of humic substances (Hansen et al., 2016). Excitation/emission pairs in the range tested here (λ_{Ex} of 420 nm, λ_{Em} between 509 and 552 nm) were also proposed elsewhere. For example, Singh and Mishra (2016) used values of λ_{Em} between 450 and 600 nm and λ_{Ex} between 375 and 550 nm for analyzing extracts from pomegranate, mixed with carbon nanoparticles.

For the aqueous solutions with fluorescein, the excitation wavelengths at maximum peak intensity of emitted light increased with Na-Fluorescein concentration from 240 to 502 nm (Table 2). The wavelengths of the emitted light at maximum peak intensity were 1.07 to 26.61 nm larger than those of the exciting light. Small differences in λ_{Ex} and λ_{Em} values (i.e., 1.07 nm) were close to the spectral resolution of the spectrometer, perhaps indicating detection of reflected light. For

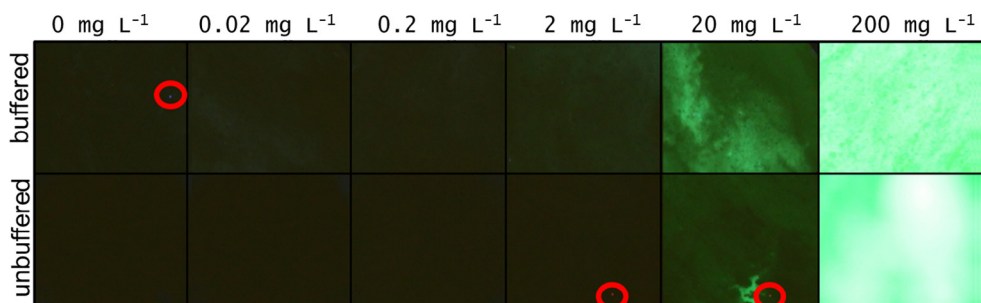


Fig. 2. Exemplary Fluorescence Images gathered within dark hood HD-50. Soil from sampling site Holzendorf, sieved to $\leq 30 \mu\text{m}$, containing buffered (top) or unbuffered (bottom) aqueous solutions with defined concentrations in Na-Fluorescein (from left to the right: 0 g L^{-1} , 0.02 mg L^{-1} , 0.2 mg L^{-1} , 2 mg L^{-1} , 20 mg L^{-1} , 200 mg L^{-1}); some dust particles appear as blue or red dots (see red circles). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 1

Mean values (with standard deviations) of color values for red, green and blue (RGB-values, Eq. (1)), as well as for the greyscale value of selected soil samples mixed with buffered or unbuffered aqueous solution containing defined Na-Fluorescein concentrations as determined by sectioning of each image.

	Concentration (mg L^{-1})	Replicate Number	Mean value (Standard deviation)				Coefficient of variation			
			Blue	Green	Red	Grey	Blue	Green	Red	Grey
buffered	0	1	37.1 (2.6)	24.3 (2.4)	0.7 (0.6)	20.2 (1.8)	7.0	9.9	79.7	8.9
	0	2	29.6 (0.6)	30.8 (0.8)	13.6 (1.7)	27.0 (0.9)	2.1	2.7	12.7	3.4
	0	3	29.6 (0.7)	30.6 (0.9)	10.0 (1.2)	26.2 (0.9)	2.4	2.9	12.0	3.3
	0	1	32.2 (0.9)	34.4 (2.0)	13.6 (2.9)	29.8 (2.1)	2.8	5.8	21.2	6.9
	0	2	32.8 (1.3)	29.7 (0.6)	5.2 (3.9)	24.7 (0.6)	4.1	2.0	75.1	2.4
	0	3	29.3 (2.3)	27.7 (1.7)	13.3 (13.2)	24.8 (3.6)	7.8	6.2	99.5	14.6
	0.2	1	31.2 (1.9)	28.9 (0.8)	8.9 (7.6)	24.8 (1.2)	6.2	2.6	85.2	4.9
	0.2	2	30.9 (1.5)	28.0 (0.7)	5.4 (5.8)	23.4 (0.8)	4.8	2.4	107.4	3.5
	0.2	3	32.3 (0.7)	33.8 (1.0)	12.2 (1.3)	29.1 (0.9)	2.3	2.8	10.5	3.2
	2	1	32.6 (0.8)	40.6 (2.8)	14.2 (2.7)	34.4 (2.6)	2.4	7.0	19.0	7.6
	2	2	30.6 (0.7)	31.7 (0.5)	3.3 (0.7)	25.6 (0.4)	2.3	1.7	20.1	1.6
	2	3	30.6 (0.7)	31.7 (0.5)	3.3 (0.7)	25.6 (0.4)	2.3	1.7	20.1	1.6
	20	1	35.0 (2.0)	80.9 (7.7)	25.8 (4.4)	65.8 (6.5)	5.7	9.5	17.0	9.9
	20	2	35.1 (1.5)	80.4 (5.2)	27.4 (3.2)	65.8 (4.4)	4.2	6.4	11.8	6.7
	20	3	37.4 (6.3)	97.9 (27.0)	32.8 (16.9)	79.7 (23.3)	16.7	27.6	51.5	29.3
	200	1	143.1 (6.0)	249.9 (6.8)	60.5 (5.0)	201.9 (6.2)	4.2	2.7	8.2	3.1
	200	2	141.6 (4.6)	249.2 (6.1)	58.6 (4.8)	200.9 (5.6)	3.2	2.4	8.1	2.8
	200	3	149.6 (10.7)	254.5 (0.1)	187.4 (8.4)	232.7 (2.6)	7.1	0.0	4.5	1.1
unbuffered	0	1	36.5 (1.4)	21.9 (0.5)	0.8 (0.8)	18.5 (0.4)	3.9	2.1	90.5	1.9
	0	2	38.5 (0.8)	28.3 (0.6)	0.6 (0.2)	23.2 (0.4)	1.9	2.0	31.3	1.9
	0	3	29.8 (0.7)	27.9 (0.3)	7.0 (1.1)	23.6 (0.3)	2.3	1.0	16.0	1.2
	0	1	30.5 (0.6)	29.1 (0.3)	8.3 (0.9)	24.8 (0.3)	2.0	0.9	11.0	1.2
	0	2	33.1 (0.7)	29.6 (0.4)	3.5 (1.0)	24.3 (0.4)	2.1	1.3	27.5	1.6
	0	3	33.2 (0.7)	29.8 (0.3)	3.5 (0.6)	24.5 (0.3)	2.0	1.0	16.9	1.1
	0.2	1	33.1 (0.8)	30.0 (0.4)	4.2 (0.6)	24.7 (0.3)	2.4	1.2	14.2	1.3
	0.2	2	32.7 (0.7)	29.6 (0.3)	3.5 (0.7)	24.2 (0.2)	2.2	0.9	18.7	0.9
	0.2	3	31.0 (0.7)	29.4 (0.6)	7.7 (0.8)	24.9 (0.6)	2.3	2.1	10.4	2.3
	2	1	31.1 (0.7)	32.3 (3.0)	7.2 (1.1)	26.9 (2.3)	2.4	9.4	15.8	8.7
	2	2	31.3 (0.6)	30.3 (0.4)	3.3 (0.6)	24.6 (0.3)	2.0	1.2	18.6	1.1
	2	3	31.0 (0.8)	29.9 (0.3)	3.2 (0.6)	24.3 (0.2)	2.4	1.1	18.8	1.0
	20	1	37.1 (1.6)	59.1 (10.3)	1.2 (1.0)	45.2 (7.6)	4.3	17.4	82.6	16.8
	20	2	35.7 (0.9)	52.5 (3.9)	0.9 (0.2)	40.3 (2.8)	2.5	7.4	27.1	6.9
	20	3	29.0 (0.8)	50.2 (4.0)	9.0 (1.2)	39.9 (3.0)	2.8	7.9	13.4	7.6
	200	1	161.4 (41.9)	254.7 (0.1)	202.6 (25.1)	236.9 (8.4)	25.9	0.0	12.4	3.5
	200	2	80.4 (4.2)	187.7 (9.6)	31.4 (4.5)	146.7 (8.1)	5.2	5.1	14.5	5.5
	200.0	3	76.7 (3.3)	183.8 (8.0)	31.2 (3.7)	143.6 (6.7)	4.3	4.4	11.7	4.7

Na-Fluorescein concentrations $\geq 0.006 \text{ mg L}^{-1}$, peak intensity was found to increase with concentration (Table 2).

The best fit between fluorescence intensity and Na-Fluorescein concentration was identified for maximum single peak intensities with λ_{Ex} equal to 420 nm and a linear regression model (Fig. 7). The λ_{Ex} value of 420 nm differed from the one λ_{Ex} that delivered the maximum peak intensity in emitted light (Table 2). The emission wavelengths emitted by Na-Fluorescein with λ_{Ex} of 420 nm were in a narrow range (509 to 552 nm, close to that measured by Gerke et al. (2013b)). Under excitation with 420 nm, the peak intensity increased with Na-Fluorescein concentration over the complete range of concentrations (Fig. 7). Compared to maximum peak intensity, the relative peak intensities (i.e., peak intensity at given λ_{Ex} divided by maximum peak intensity) ranged between 15 and 36% for excitation with 420 nm (Table 2).

The spectral evaluation as cumulated emission intensities in a range

around the emission peaks did not improve the fit of the regression between intensity and concentration (not shown). However, for Na-Fluorescein concentrations higher than those used in the presented experiments the relationship of concentrations of Na-Fluorescein and fluorescence intensity at λ_{Ex} values of 420 nm could potentially become non-linear (Gerke et al., 2013b). This is caused by reaching the detection limit of the spectrometer or by the self-quenching effect. Self-quenching effects (Lakowitz, 1999) could be indicated by a non-linear relationship between Fluorescence intensity and concentration as reported in experiments with a concentration of $> 2 \text{ mg L}^{-1}$ Na-Fluorescein (Gerke et al., 2013b). However, in the present study, the spectroscopically measured Fluorescence intensities were linearly related to Na-Fluorescein concentrations in the observed range (0–200 mg L^{-1}). Thus, self-quenching did probably not occur. Nevertheless, quenching caused e.g. by complexation could potentially influence the

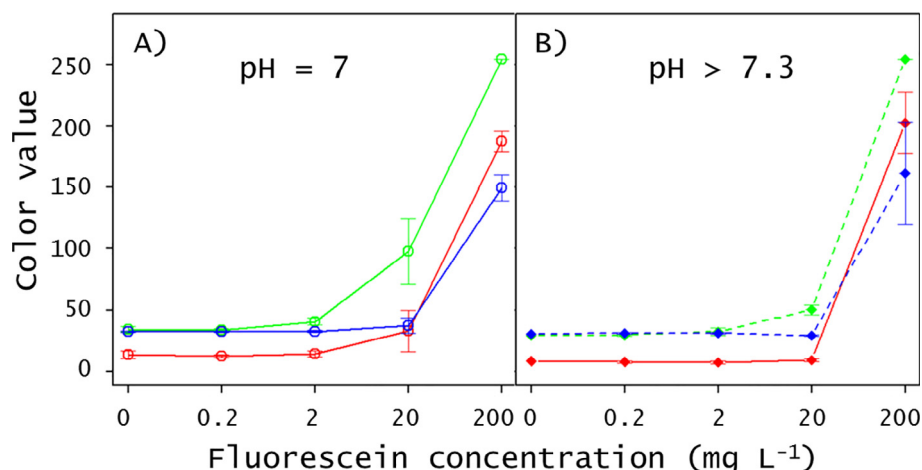


Fig. 3. Mean color values (symbols) and standard deviations (error bars) of red, green, and blue color channels of soil samples mixed with A) buffered (pH = 7) or B) not-buffered (pH > 7.3) aqueous solutions of defined Na-Fluorescein concentrations. $n = 3$ for each of the samples (i.e., three samples with un-buffered and three with buffered solution). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

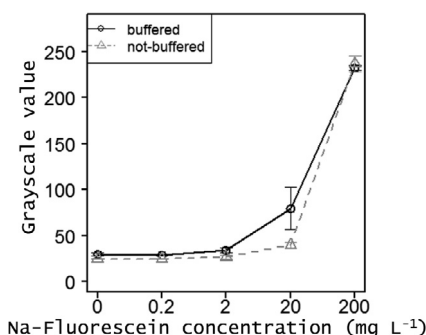


Fig. 4. Mean grayscale values (symbols) and standard deviations (error bars) as derived from color values (r, g, b; Fig. 3; using Eq. (1)) versus Na-Fluorescein concentrations. Data are from images of soil samples containing buffered (pH = 7, circles, solid line) and unbuffered (pH > 7.3, squares, dashed line) aqueous solutions with defined concentrations in Na-Fluorescein. Values close to 255 indicate the upper detection limit. $n = 3$ for each of the samples (i.e., three samples with un-buffered and three with buffered solution).

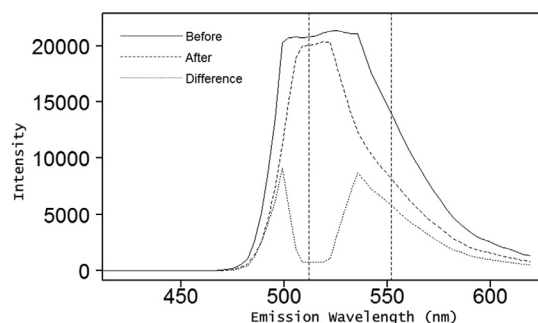


Fig. 6. Fluorescence intensity versus emission wavelengths spectra at fixed excitation wavelength of 489 nm for an aqueous Na-Fluorescein solution of a concentration of 0.4 mg L^{-1} before (solid line) and after (dashed line) filtration. The dotted line represents the difference in fluorescence intensity as caused by filtration. Vertical lines surround the range in which maximum peak emission wavelengths were identified within this study (Table 2).

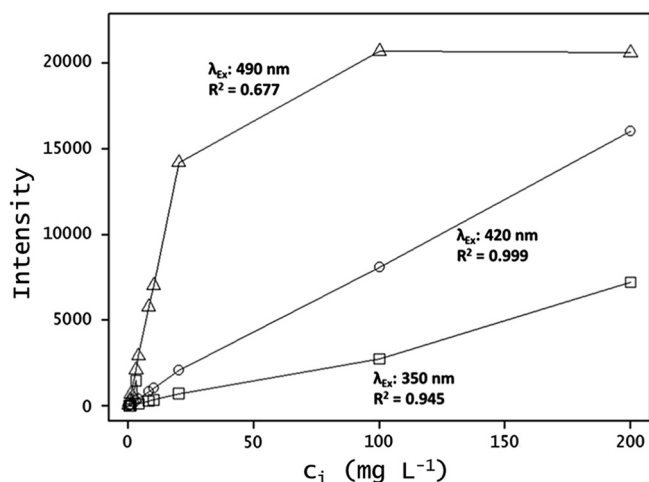


Fig. 5. Fluorescence intensity versus Na-Fluorescein concentration (c_i in mg L^{-1}) at three excitation wavelengths (490 nm, 420 nm, 350 nm) for aqueous solution samples determined with Fluorescence spectroscopy; samples were 100-times diluted.

Fluorescence intensity, but the study of this process was not in the scope here. However, effects of local variations in pH, or in the activities of other charge-determining cations that could form inner-sphere complexes should be considered in subsequent investigations.

In Fig. 7, the intercept (103.3) reflects the contribution of the

fluorescence-active organisms and other organic components such as DOM (Hansen et al., 2016).

The linear relation between fluorescence intensity and c_i (Fig. 7) was used to calculate Na-Fluorescein concentrations (mg L^{-1}) in solution (either as c_i or c_{eq}) as a function of the fluorescence peak intensity (FPI) with λ_{ex} of 420 nm as:

$$c = (FPI - 103.3)/97.98 \quad (2)$$

The addition of soil (Fig. 7) led to a decrease in fluorescence intensity as measured with Fluorescence spectroscopy for the dye-tracer solution mixed with soil. This decrease reflected the amount of dye sorbed to soil. The fluorescence peak intensity did not change for c_i between 0 and 25 mg L^{-1} , but intensity increased for c_i larger than 25 mg L^{-1} , reflecting the adsorption of Na-Fluorescein (Fig. 7). The equilibrated Na-Fluorescein concentration in the soil solution (c_{eq}) was calculated using Eq. (2). Whether c_s is plotted against c_i or against c_{eq} (Fig. 8), in both cases, the relations were linear for values of $c_i < 100 \text{ mg L}^{-1}$, also for c_{eq} considering dilution of 1:100). Here, mostly more than 60% of the fluorescein of c_i was sorbed to the soil. The sorption capacity for Fluorescein is indicated by deviation from the linearity (Fig. 8) for concentrations of $c_i > 100 \text{ mg L}^{-1}$. The adsorption isotherm (Fig. 8) was calculated under the assumption, that $c_{eq} = c_i - c_s$. This means that all the fluorescence is caused by the anion of the Na-Fluorescein although other organic compounds could be fluorescent, too (Hansen et al., 2016). Furthermore, soil properties of biopore walls could differ from those of the soil matrix (e.g., physico-chemical properties (Leue et al., 2010; Haas et al., 2018) and structural properties (Haas and Horn, 2018). This could lead to misinterpretation

Table 2

Excitation and emission wavelengths for emitted maximum peak intensity, as well as, the emitted wavelength and maximum peak intensity for excitation with $\lambda_{\text{Ex}} = 420$ nm for aqueous solutions with fluorescein in defined concentrations. Excitation⁴²⁰, Emission Peak (nm) at excitation with 420 nm for best fit as described in Section 2.3. Relative Peak Intensity: Peak Intensity at excitation with 420 nm divided by maximum peak Intensity.

Concentration	Excitation	Emission	Peak Intensity	Excitation ⁴²⁰	Peak Intensity	Rel. Peak Intensity
mg L ⁻¹	nm		–	Peak (nm)	–	
0.004	240	241.1	641	515.6	42	0.066
0.006	243	266.7	559	512.3	61	0.110
0.01	489	515.6	698	509.0	106	0.153
0.03	387	386.7	2626	512.3	310	0.118
0.04	489	512.3	2937	515.6	424	0.144
0.08	489	515.6	5806	515.6	840	0.145
0.1	489	512.3	7061	512.3	1048	0.148
0.2	489	512.3	14,207	515.6	2101	0.148
1	489	515.6	22,710	515.6	8118	0.357
2	502	525.6	24,037	552.3	16,040	0.667

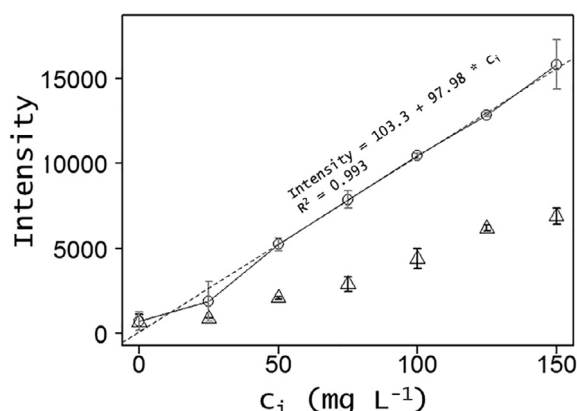


Fig. 7. Fluorescence intensity versus Na-Fluorescein concentration (mg L⁻¹) of pure aqueous solutions (c_i ; circles) and centrifuged and filtrated aqueous solutions from soil samples (c_{eq} ; triangles, $n = 3$) mixed with Na-Fluorescein solutions of defined concentrations, for $\lambda_{\text{Ex}} = 420$ nm; samples were diluted by a factor of 100.

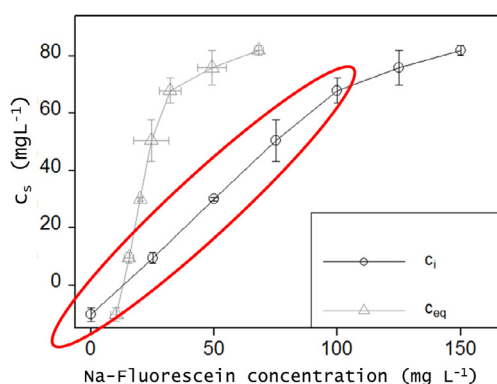


Fig. 8. Sorbed dye concentration (c_s) depending on the initial Na-Fluorescein concentration (c_i) and equilibrated Na-Fluorescein concentration (c_{eq}) in soil solution as calculated from batch experiments under the assumption $c_s = c_i - c_{eq}$. The linear range of the sorption isotherm is marked in red and described with an equilibrium partition coefficient $K_{eq} = (c_i - c_{eq}) c_{eq}^{-1}$. $n = 3$. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

of the sorbed amounts of Fluorescein, because both, the soil structure and a reduced wettability could reduce the accessibility of sorption sites, while organic compound or ions that can form inner-sphere complexes, could increase the sorption capacity locally. By sieving of the samples for preparing soil-water mixtures, the accessibility of

sorption sites can be assumed to increase because the soil structure was disturbed, coarser particles were removed and most soil organic matter remained. Thus, the sorption could perhaps be overestimated during the calibration experiments. To describe the sorption of solutes to the walls of macropores batch experiments with soil excavated from these soil volume regions is needed. In further studies locale-scale properties (e.g., cation exchange capacities, distribution and composition of organic matter, determined with DRIFTS (Leue et al., 2010, 2013; Haas et al., 2018)) should be linked to the sorption and flow of water through the macropore-matrix interface.

Another critical aspect is the incubation time. In the calibration procedure, the incubation time was 48 h, while in the (dynamic) spray experiment almost no incubation time (at the wetting front) existed. Thus, for the experiments presented in this paper, the assumption of Local Equilibrium Adsorption (LEA) of the Na-Fluorescein in soil-water suspension is critical. However, local equilibrium between the concentration of absorbate in the bulk fluid and surface phases is often assumed in the description of adsorption processes (Koopman et al., 1992). For LEA, the desorption rate equals the sorption rate. Slow rate processes such as Ostwald ripening, diffusion into micropores, re-conformation of the soil organic matter and clay-OM complexes during drying can decrease the desorption regarding the 24 h incubation time typically used in isotherm experiments without soil desiccation.

3.3. Linking fluorescence imaging with fluorescence spectroscopy

Fig. 9 shows the linkage of both techniques (Fluorescence imaging and fluorescence spectroscopy), derived from sieved samples. Two

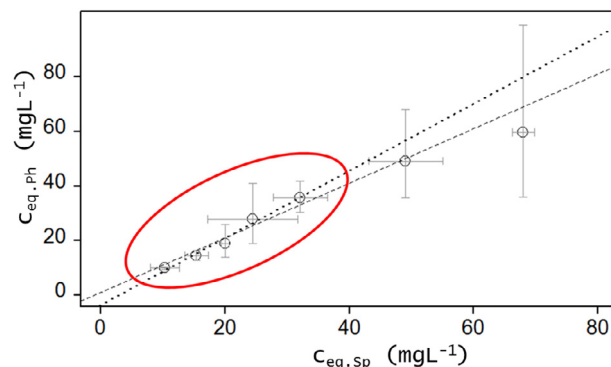


Fig. 9. Means and standard deviations of dye concentration in soil solution (with c_i : 0–100 mg L⁻¹ Na-Fluorescein) determined with fluorescence photography ($c_{eq, Ph}$) versus the concentration determined with fluorescence spectroscopy ($c_{eq, Sp}$ determined after dilution by 1:100 with distilled water). The linear range is marked in red. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

samples, with the highest fluorescein concentrations were out of the linear range of the relationship between the spectroscopic ($c_{eq,Sp}$) and photo-based derived fluorescein concentration ($c_{eq,Ph}$). This is caused by reaching the upper detection limit of the cmos sensor ($c_{eq,Ph}$). The transferability, between the two methods, is indicated by the regression line (Fig. 9), which (for $c_i < 100 \text{ mg L}^{-1}$) is as follows:

$$c_{eq,Sp} = -3.58 + 1.29c_{eq,Ph} \quad (3)$$

where $c_{eq,Ph}$ is Fluorescein concentration in soil solution determined with Fluorescence imaging (mg L^{-1}) and $c_{eq,Sp}$ is the concentration in solution as determined with Fluorescence Spectroscopy (mg L^{-1}).

For aqueous solutions mixed with soil, the decrease in fluorescein intensity (i.e., determined with *fluorescence spectroscopy*) was used to determine the sorption of fluorescein to the soil (Fig. 7). This is exemplarily shown for the HD1 sample from Holzendorf. First, $c_{eq,Ph}$ (i.e., the “corrected” fluorescein concentration in the soil solution, responsible for fluorescence in fluorescence images) was calculated from mixed samples and c_i with the help of the adsorption isotherm (Fig. 8), assuming that $c_{eq,Sp} = c_{eq,Ph} = c_i - c_s$. Once, the “corrected” fluorescein concentration is known, $c_{eq,Ph}$ can be derived from grayscale values of fluorescence imaging with the help of Eq. 4 which is derived from data shown in Fig. 10:

$$c_{eq,Ph} = a * e^{b*GS} \quad (4)$$

with a equal to 7.72 and b equal to 0.0118 as fitting parameters, grayscale value (GS) and e is Euler's number. The 95% confidence interval (Fig. 10) revealed increasing experimental uncertainty with grayscale values. This could be caused by the relatively small number of replicates ($n = 3$), by relatively small range of observed grayscale values at the lower concentrations (Fig. 10), and methodological issues such as spatially heterogeneous soil properties.

3.4. Fluorescence imaging on an intact soil aggregate

The resulting fluorescence image for an aggregated soil sample is shown in Fig. 11. With imaging in UV, equilibrated Na-Fluorescein concentrations (c_{eq}) of about 60 mg L^{-1} (red color in Fig. 11) were found at the surfaces of the sprayed area and within the first millimeters in distance from the interface. This concentration equals c_i (i.e., 100 mg L^{-1}) used for spraying. Intermediate concentrations of $15\text{--}20 \text{ mg L}^{-1}$ (green and yellow color) can be found within the next centimeter and appear in a band, parallel to the sprayed interface. Most of the soil is low in Na-Fluorescein concentration ($< 12 \text{ mg L}^{-1}$, blue colored). Spots of higher Na-Fluorescein concentrations at some distance from the application surface represent flow paths with rapid water movement (marked with red circles in Fig. 11). At one of these spots, a root channel was identified within the image. The spots appear

to be completely isolated from the surrounding and disconnected from the sprayed clod surface because the 3D connecting pore network is not in the layer of the soil slice, from which the photo was taken. The high spatial resolution enabled to identify the effects of such local or pore-scale flow paths, which seem to impact the exchange between preferential flow paths and the soil matrix also at the macroscopic scale. Further investigations are required to account for these structural elements in soils, their functioning in the plant-soil-atmosphere continuum and their management induced alterations. The sorbed dye concentrations are mapped in Fig. 12. Again, the highest concentrations are found with the interface, while isolated spots high in sorbed dye concentrations are identified. However, by applying the calibration procedure to soil from the soil matrix and to soil of the interfaces, visualization and quantification of mass exchange between preferential flow paths and the soil matrix may be further improved.

4. Conclusions

The study aimed at developing an approach for the experimental determination of sorption effects during preferential flow in structured soil. The idea was to quantify tracer distributions from photo images in order to trace the movement of a reactive solute from the surface of an aggregate into the soil matrix. The focus here was on testing the suitability of Na-Fluorescein and a photographic mapping of tracer distributions.

The calibration results suggest that Na-Fluorescein is useful for mapping the effect of small-scaled distributions of sorption properties at intact aggregate surfaces and of the fluorescein concentrations in slices of the soil matrix from photo-based image analyses. A linear relationship between fluorescence intensity and concentration in solution, determined using a spectrometer, could be established by mass balancing with the absorbed amount of fluorescein. The photo-image results suggest that a high spatial resolution of up to $69 \mu\text{m}$ pixel size is possible. The resolution of the mapped concentrations in the aqueous phase depends on the proper calibration.

When considering that the fluorescence intensity of absorbed fluorescein is negligibly small, the spatial distribution of both, the dissolved and the absorbed fluorescein concentrations can be quantified from the images.

For the present concentration range, the spectroscopic analyses revealed that an excitation wavelength of 420 nm was superior to previously suggested values of 350 and 490 nm . Fluorescence imaging showed that sorption of the dye mostly occurs directly along the aggregate surface. Spots high in concentration indicated the occurrence of rapid movement along root channels suggesting a secondary preferential flow and transport mechanism at the aggregate scale.

This approach can be useful when trying to obtain data (i.e., time series' of concentration maps) for the numerical simulation of 2D or 3D small-scaled water and solute movement in the soil matrix during preferential flow along macropore surfaces and for the analysis of macropore-matrix reactive solute mass exchange in two-domain models.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.geoderma.2019.114002>.

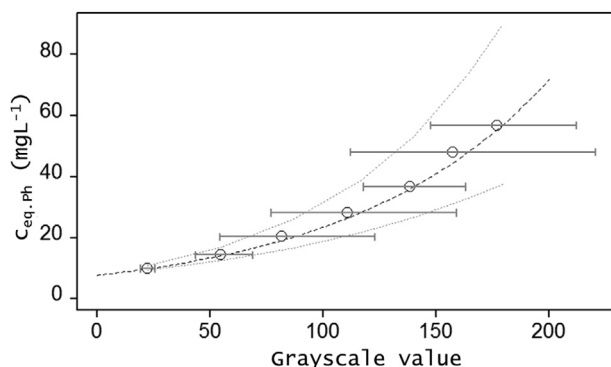


Fig. 10. Means and 95% confidence interval of equilibrated Na-Fluorescein concentration ($c_{eq,Ph}$) as a function of grayscale values (circles: means with standard deviations). The dotted line represents the used model shown in Eq. (4).

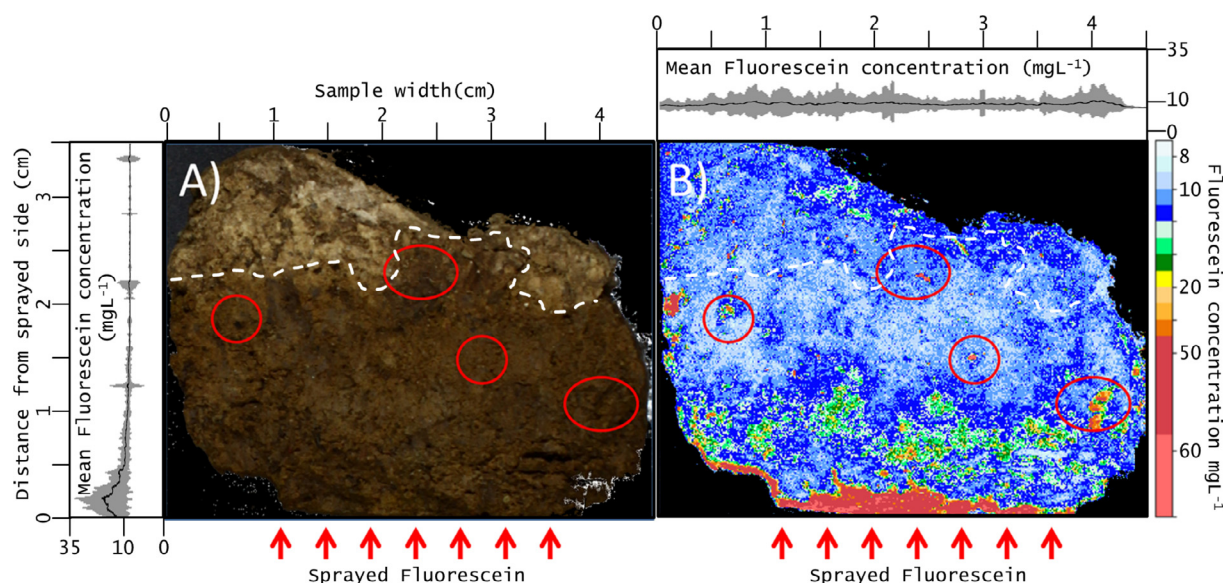


Fig. 11. A) Plan-view photo and B) calculated fluorescein concentrations of an aggregated soil sample (i.e., a clod of 4–5 cm edge length). At the sprayed side of the aggregate (bottom part) the macropore wall regions with almost initial Na-Fluorescein concentrations ($c_i = 100 \text{ mg L}^{-1}$) are indicated by red colors. In most of the soil matrix inside the clod, the concentration was $< 6 \text{ mg L}^{-1}$. Specific regions with high Na-Fluorescein concentration are marked with red circles (from left to right: a vertical pinhole, a root, another vertical pinhole, and a horizontal biopore). Dotted-line: Wetting front. The brighter shining soil remained air-dried at greater distances from the sprayed side beyond the wetting front. The sample was sliced and photographed after 3.5 h of spraying of the tracer solution at a rate of 0.9 mg s^{-1} . (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

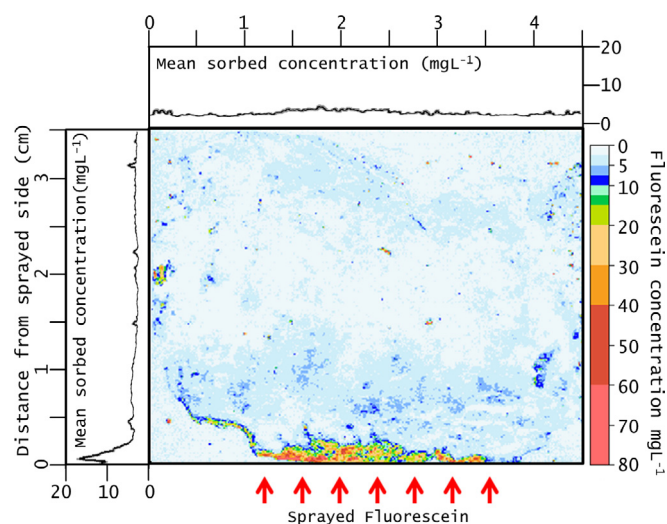


Fig. 12. Sorbed dye concentrations as calculated for the aggregated soil sample, shown in Fig. 11. Plots at the left-hand side and on top show means of Na-Fluorescein concentrations (black line) with standard errors (gray shadow) sorbed to the soil.

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